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CSCI 112 Final Project Video Game Sales

Case Background

Video games have become a popular pastime for different generations around the world. Its popularity has led to the rise of video game distribution stores such as GameStop and DataBlitz. These stores have allowed gamers to purchase physical copies of their favorite or the newest video games. However, digital video game stores such as Steam and Epic Games have given gamers a more convenient way to purchase video games.

This leads to the case of a new physical game store that needs to determine the optimal distribution of video games that it must sell. This store wants to ensure that it can sell video games that are popular with gamers. Yet, the owners are not sure which specific metrics they can base off so they can purchase specific video games that they can be confident that they can sell to gamer customers.

With this, the project team will act as consultants for the game store to help its owners determine what video game stock-keeping units to purchase and eventually continue distributing to ensure maximum sales for the store.

About the Dataset:

This [Video Game Sales](#) contains sales data for over 16,500 video games across various platforms and regions. It includes game titles, genres, platforms, release years, and sales figures in North America, Europe, Japan, and other countries.

- Rank - Ranking of overall sales
- Name - The game's name
- Platform - Platform of the game's release (i.e., PC, PS4, etc.)
- Year - Year of the game's release
- Genre - Genre of the game
- Publisher - Publisher of the game
- NA_Sales - Sales in North America (in millions)
- EU_Sales - Sales in Europe (in millions)
- JP_Sales - Sales in Japan (in millions)
- Other_Sales - Sales in the rest of the world (in millions)
- Global_Sales - Total worldwide sales.
- Critic_score - Aggregate score compiled by Metacritic staff
- Critic_count - The number of critics used in coming up with the Critic_score
- User_score - Score by Metacritic's subscribers
- User_count - Number of users who gave the user_score

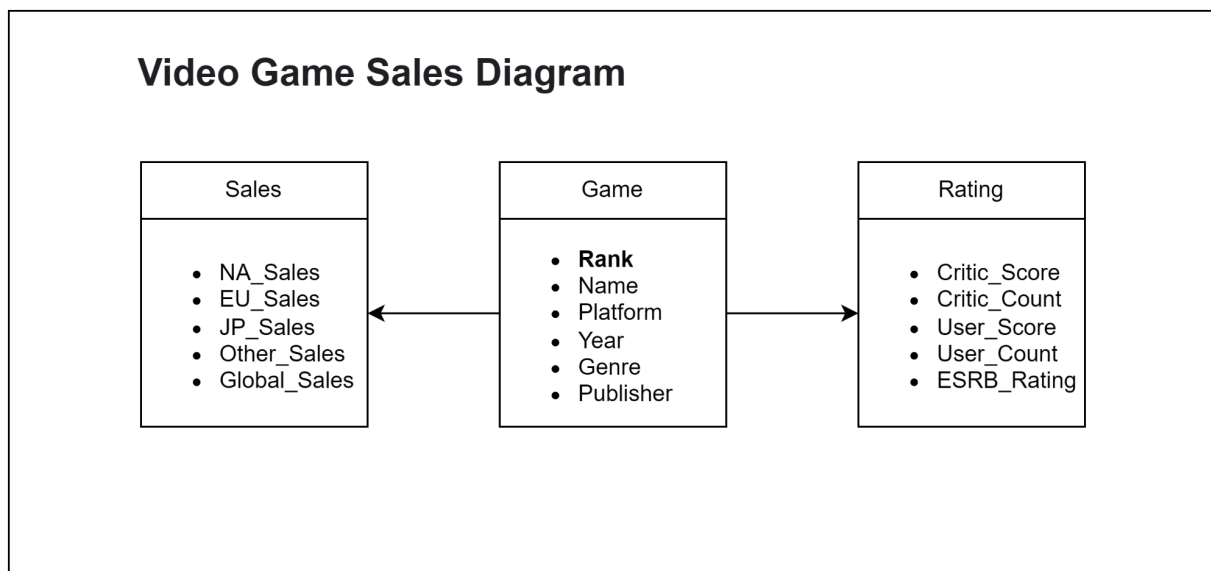
- Developer - The Party responsible for creating the game
- Rating - The ESRB ratings

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
	Name	Platform	Year_of_Release	Genre	Publisher	NA_Sales	EU_Sales	JP_Sales	Other_Sales	Global_Sales	Critic_Score	Critic_Count	User_Score	User_Count	Developer	Rating
1	Wii Sports	Wii	2006	Sports	Nintendo	41.36	28.96	3.77	8.45	82.53	76	51	8	322	Nintendo	E
2	Super Mario Bros.	NES	1985	Platform	Nintendo	29.08	3.58	6.81	0.77	40.24						
3	Mario Kart Wii	Wii	2008	Racing	Nintendo	15.68	12.76	3.79	3.29	35.52	82	73	8.3	709	Nintendo	E
4	Wii Sports Resort	Wii	2009	Sports	Nintendo	15.61	10.93	3.28	2.95	32.77	80	73	8	192	Nintendo	E
5	Pokemon Red/Pokemon Blue	GB	1996	Role-Playing	Nintendo	11.27	8.89	10.22	1	31.37						
6	Tetris	GB	1989	Puzzle	Nintendo	23.2	2.26	4.22	0.58	30.26						
7	New Super Mario Bros. DS	DS	2006	Platform	Nintendo	11.28	9.14	6.5	2.88	29.8	89	65	8.5	431	Nintendo	E
8	Wii Play	Wii	2006	Misc	Nintendo	13.96	9.18	2.93	2.84	28.92	58	41	6.6	129	Nintendo	E
9	New Super Mario Bros. Wii	Wii	2009	Platform	Nintendo	14.44	6.94	4.7	2.34	28.32	87	80	8.4	594	Nintendo	E
10	Duck Hunt	NES	1984	Shooter	Nintendo	26.93	0.63	0.28	0.47	28.31						
11	Nintendogs	DS	2005	Simulation	Nintendo	9.05	10.95	1.93	2.74	24.67						
12	Mario Kart DS	DS	2005	Racing	Nintendo	9.73	7.47	4.43	1.8	33.43	84	64	8.8	484	Nintendo	E

Data Model Design:

The following diagram shows how we organized the dataset. We chose to model the video game data as a single collection where each document represents one game, and rank is considered the primary key along with the provided ID by MongoDB. The sales subdocument groups all sales-related fields (na, eu, jp, other, global) to improve organization. This action was also done for the rating subdocument for all rating-related fields (critic_score, critic_count, user_score, user_count, esrb_rating).

The model will be used for MongoDB, as all data about a game is stored in one place. This structure allows fast aggregation queries (e.g., total sales by genre, platform, or publisher).



Primary Key	Attributes						
Rank	Name	Platform	Year	Genre	Publisher	Sales	Ratngs
1	Wii Sports	Wii	2006	Sports	Nintendo	{ sales: { na: 41.49 eu: 29.02 jp: 3.77 other: 8.46 global: 82.74 } }	{ sales: { critic_score: 51.0 critic_count: 76.0 user_score: E user_count: 322.0 esrb_rating: 8.0 } }
2	Super Mario Bros.	NES	1985	Platform	Nintendo	{ sales: { na: 29.08 eu: 3.58 jp: 6.81 other: 0.77 global: 40.24 } }	{ sales: { critic_score: nan critic_count: nan user_score: nan user_count: nan esrb_rating: nan } }

Data Preparation:

The group created 'import_vgsales.py', where we imported and used the Pandas library to preprocess and clean the data. We then converted it to a list of dictionaries, since this is what MongoDB reads to import the data into the cloud. The photo below shows the first 2 documents from the collection 'games'.

```
{ '_id': ObjectId('6810bc9c80a4e749f058bffb'),
  'developer': 'Nintendo',
  'genre': 'Sports',
  'name': 'Wii Sports',
  'platform': 'Wii',
  'publisher': 'Nintendo',
  'ratings': {'critic_count': 51.0,
              'critic_score': 76.0,
              'esrb_rating': 'E',
              'user_count': 322.0,
              'user_score': 8.0},
  'sales': {'eu': 28.96,
            'global': 82.53,
            'jp': 3.77,
            'na': 41.36,
            'other': 8.45},
  'year_of_release': 2006.0}
{ '_id': ObjectId('6810bc9c80a4e749f058bffc'),
  'developer': nan,
  'genre': 'Platform',
  'name': 'Super Mario Bros.',
  'platform': 'NES',
  'publisher': 'Nintendo',
  'ratings': {'critic_count': nan,
              'critic_score': nan,
              'esrb_rating': nan,
              'user_count': nan,
              'user_score': nan},
  'sales': {'eu': 3.58, 'global': 40.24, 'jp': 6.81, 'na': 29.08, 'other': 0.77},
  'year of release': 1985.0}
```

Queries:

1. Top 10 Best Selling Platforms Globally

The query works by grouping all the video games by platform (Wii, PS4, X360, etc) and then summing up their global sales. It's then sorted in order to get the top 10. This query can help a store identify which platforms are most popular, which will allow them to make more informed decisions when it comes to inventory prioritization, promotions, and investing in accessories and games for those platforms.

2. Top 10 Best Selling Publishers Globally

The query groups the games by publisher (EA, Nintendo, etc) and like the previous query, totals the global sales then sorts them to get the top 10. This can help the store identify which publishers are consistently selling, which can allow the store to feature these publishers more prominently and even plan featured publisher events to maximize sales.

3. Best-Selling Game per ESRB Rating

The query returns the top selling game by global sales for each ESRB rating (E, T, M, etc). This helps the store stay informed on what kinds of games sell in each age group. It will help them tailor marketing and target specific age demographics better.

4. Average Critic Score vs User Score by Genre (minimum 5 games)

This query gets the average critic and user scores for each genre, making sure that the genre has at least 5 games to be fair. This can show which genres rate consistently well for both critics and players alike, which can be a good guide for stores to see which genres have long-term sales potential, as games that are well-rated and received tend to be easily sold past the peak of their popularity.

5. Best-Selling Video Game Each Year in Terms of Global Sales

The query shows the top-selling game of each year in terms of global sales. This will allow the store to be informed and leverage "Game of the Year" type marketing for these extremely popular games. Aside from this, it offers some insight into the yearly changes and trends.

6. Most Successful Genre per ESRB Rating Based on Average User Score

The query shows the top rated genre based on average used score per ESRB rating, again considering genres with at least 5 games for fairness. Similar to the third query, this will inform the store of age targeted recommendations, as well as genre focused marketing, highlighting certain genres for specific age demographics.